



UNIVERSITY OF THE FREE STATE

BLOEMFONTEIN CAMPUS

CSIS3744
Memorandum

**DEPARTMENT OF COMPUTER SCIENCE AND
INFORMATICS**

CONTACT NUMBER: 051 401 3702

EXAMINATION: Additional End-of-Year 2020

ASSESSOR: 1. Prof Okuthe P. Kogeda

MODERATOR: 1. Mnr. W.S.J. Marais

TIME: 3 HOURS

MARKS: 100

INSTRUCTIONS: Answer ALL the questions. Write legibly. Number your questions clearly and correctly.

Question 1 [24 Marks]

Dwesa resort wants to divide their network into two subnets. They also want to implement multiple virtual servers on a single host machine. Their current network address is 192.168.5.0.

- (a) To which network class does the above mentioned network address belong? 1 Mark]

<<< Class C >>>

- (b) Calculate the subnet mask for Dwesa resort. [3 Marks]

<<< Default subnet mask for class C network is 255.255.255.0.

Binary value of subnet mask is 11111111.11111111.11111111.00000000

$$2^n - 2 = 2 \quad n = 2 \quad \checkmark$$

Change 2 bits from 0 to 1 for the last octet.

11111111.11111111.11111111.11000000

Value of 11000000 $\rightarrow 2^7 + 2^6 = 128 + 64 = 192 \quad \checkmark$

The new subnet mask is 255.255.255.192 $\checkmark$

OR

Default subnet mask for class C network is 255.255.255.0

Binary value of subnet mask is 11111111.11111111.11111111.00000000

$$2^n = 2 \quad n=1 \quad \checkmark$$

Change 1 bit from 0 to 1 for the last octet.

11111111.11111111.11111111.10000000 $\checkmark$

Value of 10000000 $\Rightarrow 2^7 = 128$

The new subnet mask is 255.255.255.128 $\checkmark >>>$

- (c) How many machines will each subnet be able to host? [3 Marks]

<<< $(2^6 - 2) \checkmark = 62 \quad \checkmark \checkmark$

OR

$(2^7 - 2) \checkmark = 126 \quad \checkmark \checkmark >>>$

- (d) Complete the table below by supplying each subnet's network address, broadcast address and usable host address range: [4 Marks]

	Subnet	Network address	Broadcast address	Usable host address range
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(i)	1	192.168.5.64 ½	192.168.5.127 ½	192.168.5.65 ½ to 192.168.5.126 ½
(ii)	2	192.168.5.128 ½	192.168.5.191 ½	192.168.5.129 ½ to 192.168.5.190 ½

OR

	Subnet	Network address	Broadcast address	Usable host address range
(i)	1	192.168.5.0 ½	192.168.5.127 ½	192.168.5.1 ½ to 192.168.5.126 ½
(ii)	2	192.168.5.128 ½	192.168.5.255 ½	192.168.5.129 ½ to 192.168.5.254 ½

- (e) Explain the disadvantages that the implementation of multiple virtual servers on a single host machine might hold. [4 Marks]

<<< Compromised performance: When multiple VMs contend for finite physical resources, one VM could monopolize those resources and impair the performance of other VMs on the same computer. ✓ Increased complexity: Although virtualization reduces the number of physical machines to manage, it increases complexity and administrative burden in other ways. ✓ Increased licensing costs: Because every instance of commercial software requires its own license, every VM that uses such software comes with added cost. ✓ Single point of failure: If a host machine fails, all its guest machines will fail too. ✓ >>>

- (f) Briefly explain the difference between a bridged, NAT and host-only network connection type with regard to virtualization. [3 Marks]

<<< Bridged: The vNIC accesses the physical network using the host machine's NIC and obtains its own IP address, default gateway, and subnet mask from the DHCP server on physical LAN. ✓

NAT: The virtual network is hidden from the physical network but can still communicate with it. It obtains its IP address information from the virtual DHCP server on the host machine. ✓

Host-only: VMs on one host can exchange data with each other and the host but cannot communicate with nodes beyond the host. ✓ >>>

(g) What are the benefits of VLANs?

[3 Marks]

<<< VLANs allow the creation of collision domains by groups other than just physical location. ✓ Using VLANs, it is possible to establish networks by different means, such as by function, type of hardware, protocol, among others. ✓ This is a big advantage when compared to conventional LANs wherein collision domains are always tied to physical location. ✓ >>>

(h) Why is network segmentation a good idea when managing a large network? [3 Marks]

<<< For a network administration, segmenting a network would help ease network traffic ✓ and ensures that high bandwidth is made available at all times for all users. ✓ This translates to better performance especially for a growing network. ✓ >>>

Question 2 [44 Marks]

You have been employed by University of the Free State Enterprises (UFSE) as network specialist. This is the arm of the university that deals with commercialization of various ideas developed by the academic staff members and students. The main office is in Bloemfontein where data is stored and some processing take place and they have decided to open a branch office in Johannesburg, which is near to most of their clients. The branch office is set to start with 10 employees with a view of expansion up to 50 employees in future. However, due to cost of setting up the network, UFSE wants to have 6 work stations in the office, and the remaining employees will use tablets or laptops since they are on the move as marketers, salespersons, and support staff to connect wirelessly. You have been assigned the first task to design and implement a network for the branch in Johannesburg.

(a) State the purpose of cables being shielded and having twisted pairs.

[2 Marks]

<<< The main purpose of this is to prevent crosstalk. Crosstalks are electromagnetic interferences or noise that can affect data being transmitted across cables. ✓✓ >>>

(b) State the danger of using cables longer than the prescribed length.

[2 Mark]

<<< Cables that are too long would result in signal loss. ✓ This means that data transmission and reception would be affected, because the signal degrades over length. ✓>>>

- (c) State why you would prefer STP to UTP in your cabling. [3 Marks]

<<< 1) STP is noise resistant. ✓ 2) STP provides better throughput. ✓ 3) STP is of high quality. ✓>>>

- (d) In doing a final configuration and testing of the network implemented, you realize signal attenuation issues. Discuss the measures you will take to solve the issue. [3 Marks]

<<< A common way of dealing with such a problem is to use repeaters and hub, because it will help regenerate the signal and therefore prevent signal loss. Checking if cables are properly terminated is also a must. ✓✓✓>>>

- (e) State the topology you would implement for the network. Discuss three reasons for your choice. [4 Marks]

<<< Star topology or bus topology or both star & bus (hybrid) topology is expected to be implemented. ✓

The advantages of either would suffice, which may include:

Star – Easy to install and wire; No disruptions to the network when connecting or removing devices; Requires more cable length than a linear bus topology; If the connecting network device (network switch) fails, nodes attached are disabled and cannot participate in computer network communication.

Bus - Easy to connect a computer or peripheral to a linear bus; Requires less cable length than a star topology; Entire network shuts down if there is a break in the main cable; Difficult to identify the problem if the entire network shuts down; Not meant to be used as a stand-alone solution. ✓✓✓>>>

- (f) Draw the physical topology of the network chosen in question 2 (e). [6 Marks]

<<< see fig. attached. >>>

- (g) Discuss the purpose of implementing a demilitarized zone (DMZ) and provide two possible types of servers you would place in the DMZ. [4 Marks]

<<< The DMZ is less secure than the internal LAN, since the external firewall is more porous than the internal firewall. ✓ This allows users from the Internet to get access to content and services on servers inside the DMZ such as web servers ✓ and external DNS ✓, while the internal LAN remains protected. ✓>>>

- (h) Explain any three transmission flaws that may affect the network. [3 Marks]

<<< The explanation of any three earns a candidate 3 marks.

1. Noise - any undesirable influence degrading or distorting the signal, measured in decibels, i.e., EMI, cross-talk, etc. ✓ 2. Attenuation - loss of signal strength as it travels away from the source. ✓ 3. Latency - the delay between signal transmission and receipt, which may cause transmission errors. It is caused by cable length, intervening connectivity device, etc. ✓ >>>

- (i) The network system just like any other system is subject to failure. Name and explain FOUR command line tools for troubleshooting network problems. [6 Marks]

<<< Candidates are supposed to get ½ mark for each 4 names provided. One mark for explanation of each 4 command line tools. The 4 tools may include: 1. Ping½ – used to verify if TCP/IP is installed, bound to NIC, configured correctly and communicating with network. ✓ 2. Ipconfig½ – shows current TCP/IP addressing & domain name information on a windows computer. ✓ 3. Ifconfig½ – to view and manage TCP/IP settings. ✓ 4. ½ Tracert – used to trace the TCP/IP on the network. ✓ 5. Nslookup½ – allows you to query the DNS database from any computer on a network to find the host name of a device and verify if its configured correctly., ✓ etc. >>>

- (j) State the reasons for implementing a DHCP server at the branch. [3 Marks]

<<< (1) Reliable IP address configuration. ✓ (2) DHCP minimizes configuration errors caused by manual IP address configuration, such as typographical errors, or address conflicts caused by the assignment of an IP address to more than one computer at the same time. ✓ (3) Reduced network administration. ✓ >>>

- (k) Discuss the physical security precautions you would consider. [4 Marks]

<<< Some of the physical security precautions include: locking of the server and data rooms; ✓ know the best escape routes or emergency exits; ✓ electrical and tool safety; ✓ materials safety; ✓ etc. >>>

- (l) In implementing the network, you considered peer-to-peer and client/server network models. You finally settled on client/server network model because of web, email, Telnet, FTP, etc. services that are going to be offered by UFSE. Explain any four additional reasons for this choice. [4 Marks]

<<< The candidates are expected to provide advantages relative to peer-to-peer networks including: 1. User credential assigned from one place ✓ 2. Multiple shared resource access centrally controlled ✓ 3. Central problem monitoring, diagnostics, and correction capabilities ✓ 4. More scalable. ✓ >>>

Question 3 [14 Marks]

TCP and UDP are major protocols used on the transport layer of the OSI model.

- (a) Why is UDP least favoured when compared to TCP? [4 Marks]

<<< It's because UDP is unreliable ✓ and un-sequenced. ✓ It is not capable of establishing virtual circuits ✓ and acknowledgements. ✓>>>

- (b) You realized that the network's performance has degraded as a result of the use of TCP on the transport layer. You consult and you are advised to change to UDP. Explain how this change would improve the performance of the network? [3 Marks]

<<< Candidates are expected to state that: UDP adds less payload (8 bytes while TCP adds 20 bytes); ✓ UDP is connectionless oriented; ✓ UDP has no checksum for errors; ✓ etc. >>>

- (c) Which fields are modified in IPv4 and IPv6 packets to help prioritize traffic for QoS? [2 Marks]

<<< DiffServ for IPv4, ✓ Traffic Class for IPv6. ✓>>>

- (d) What are some basic ways to speed up network performance? [4 Marks]

<<< There are several ways to improve network performance: You can first make sure you disable peer-to-peer downloading and torrents. ✓ Some companies restrict media streaming such as YouTube and Pandora. ✓ You can also add compression to files to reduce the amount of bandwidth used. ✓ Another area where some network admins make mistakes is timed backups and large data transfers. ✓ Make sure your backups are done during slow network times, which is usually overnight when the office is closed. ✓>>>

- (e) Which protocol can provide VoIP services over a Metro Ethernet connection? [1 Mark]

<<< SIP ✓>>>

Question 4 [18 Marks]

There are three primary areas or classifications of security controls. These include management security, operational security, and physical security controls.

- (a) Why do network administrators create domain groups to manage user security privileges? [1 Mark]

<<<To simplify the process of granting rights to users. ✓>>>

- (b) What type of scanning might identify that Telnet is running on a server? [1 Mark]

<<< Port scanning. ✓>>>

- (c) Why is a BPDU filter needed at the demarc? [2 Marks]

<<< The ISP's STP-related topology information shouldn't be mixed with a corporate network's STP-related topology information. ✓✓>>>

- (d) What characteristic of ARP makes it particularly vulnerable to being used in a DoS attack? [2 Marks]

<<< ARP performs no authentication. ✓✓>>>

- (e) What are the four phases in the social engineering attack cycle? [4 Marks]

<<< 1) Research. ✓ 2) Building trust. ✓ 3) Exploit. ✓ 4) Exit. ✓>>>

- (f) Explain two security benefits of using virtualization on a network. [2 Marks]

<<< Fault and threat isolation; ✓ Simple backups, recovery, and replication. ✓>>>

- (g) Explain four functions that all routers perform? [4 Marks]

<<< 1. Connect dissimilar networks. ✓ 2. Interpret Layers 3 and 4 addressing and other information. ✓ 3. Determine the best path for data to follow. ✓ 4. Reroute traffic if a primary path is down but another path is available. ✓>>>

- (h) Explain the difference between passive scanning and active scanning during the association process. [2 Marks]

<<< In passive scanning, a wireless station listens on all channels within its frequency range for a special signal, known as a beacon frame, issued from an access point. ✓

In active scanning, the station transmits a special frame, known as a probe, on all available channels within its frequency range. ✓ >>>

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